**NCR#: N**26436

## IDENTIFICATION

| | |
|---|---|
| Initiator: Simin Mohammad | Date: 21-Aug-2026 |
| Department: Microbiology | Sample Name/Material Description: Active Air Sample and Surface Sampling |
| Equipment ID #: Cleanroom Suite 114 (Anteroom & Buffer Room) | |
| Sample ID #: N/A 21Aug26 SMO | Sample/Material Lot #: N/A 21Aug26 SMO |

## NONCONFORMITY DESCRIPTION (Who, What, When, Where, Why, & How)

**Nonconformance Description**:

It was identified that EM analyst SMO did not perform the routine required weekly surface sampling and active air sampling for Cleanroom Suite 114 (Anteroom 114A and Buffer Room 114B), as specified in MICRO-SOP-2 (Environmental Monitoring of the Clean Room Facility), during the period of 03-Aug-2026 through 14-Aug-2026. Consequently, weekly EM data for Cleanroom Suite 114 were unavailable for two consecutive weeks.

Three (3) Celsis sterility test OOS investigations—OOS-261824 (Room 114B), OOS-261861 (Room 114B), and OOS-261877 (Room 114B)—were identified during this period. (Note: OOS-261878 and OOS-261932 were reviewed and confirmed to have been tested in Cleanroom Suite 115 and are excluded). During this two-week timeframe, 477 sterility samples were tested in Suite 114 with 474 samples passing all acceptance criteria (>99.3% pass rate), demonstrating that no widespread positive or systemic contamination trend occurred. In-process session-specific ISO 5 BSC environmental monitoring (0 CFU settling plates, acceptable contact plates, passing gloved touch plates), continuous 24/7 Lighthouse HVAC differential pressure/temp logs, and method negative controls all remained strictly within specification. Based on the lack of widespread positive results and intact primary containment, the missed weekly background EM is determined to be unrelated to these isolated sterility test failures. See attached Risk Assessment Addendum.

## CAUSE(S) ANALYSIS

**Description of cause(s)**: Analyst oversight

**Category** (*select all that apply*): ☐ Equipment/System ☐ Process/Method ☑ Personnel ☐ External Phenomena
☐ Other: N/A 21Aug26 SMO

## INTERIM CORRECTIVE ACTION(S)

**Description of corrective action(s):**

The microbiology lab supervisor instructed the analysts to be more vigilant during the environmental monitoring process to ensure that the surface sampling of the anteroom and buffer room is performed as required. On 21-Aug-2026, Microbiology Laboratory Supervisor Robin Seymour instructed the microbiology team to have actively update supervisor and lead analysts every week going forward to ensure rigorous tracking, accountability, and verification of all cleanroom sampling points. Analyst SMO will need to complete Read & Understood retraining on MICRO-SOP-2.

## QUALITY EVALUATION

**Was action effective in addressing the root cause of the nonconformity?** ☑ Yes ☐ No ☐ N/A

CAR required: ☐ YES ☐ NO *If yes, CAR #:* ___________

If yes, explain how effectiveness was determined:

**Additional Comments:** Effectiveness confirmed via immediate reinstatement of weekly EM with acceptable results and launch of Weekly EM Tracking Sheet.

Risk Assessment / Comments: Impact evaluated for 3 correlated Celsis sterility OOSs in Room 114B (OOS-261824, OOS-261861, OOS-261877). Over 470+ concurrent samples tested negative (>99.3% pass rate). Continuous 24/7 Lighthouse HVAC logs, session-specific ISO 5 BSC EM, and negative controls confirmed operational control. See attached Risk Assessment Addendum.

## ACKNOWLEDGEMENT

| | | |
|---|---|---|
| **Prepared By** (*Print*): Simin Mohammad | ______________ Signature | ______ Date |
| **Reviewed By** (*Print*): Qiyue Chen | ______________ Signature | ______ Date |

**NCR#: N**26436

**CLOSING THE NONCONFORMANCE** *(Closure by Quality)*

Assessment Impact:

Frequency: 2_________ <2 = 1 Between 2 to 10 = 2 >10 = 3

Severity: 2____________

Typo Corrections = 1
Changes in processes and/or systems with no retroactive implications = 2
Changes in processes and/or systems with retroactive implication =3

Magnitude: 2___________

<2 Process/Material/Equipment Affected = 1
>2 to <10 Process/Material/Equipment Affected = 2
>10 Process/Material/Equipment = 3

1 to 2: Low
3 to 17: Major
18 to 27: Critical

| | | |
| --- | --- | --- |
| 3 | 18 | 27 |
| 2 | 17 | 18 |
| 1 | 2 | 3 |

Assessment Impact Grading: 8

Impact Level

☐ Low ☑ Major ☐ Critical

| **Quality Reviewed By** *(Print)*: | ______________________ Signature | ________ Date |
| --- | --- | --- |
| **Quality Approved By** *(Print)*: | ______________________ Signature | ________ Date |

# QUALITY RISK ASSESSMENT & PRODUCT IMPACT EVALUATION

## Addendum to Nonconformance Report NCR# N26436

| | | |
|---|---|---|
| **NCR Number:** N26436 | **Date of Evaluation:** 26-Aug-2026 / 27-Aug-2026 | **Department:** Microbiology (Sterility) |
| **Affected Location:** Cleanroom Suite 114 (Anteroom 114A & Buffer Room 114B) | **Missed Period:** 03-Aug-2026 to 14-Aug-2026 (2 Weeks) | **Applicable SOPs:** MICRO-SOP-2, CORP-SOP-4, SOP 2.100.019 |

### 1. Executive Summary & Nonconformance Scope

During the supervisory review of environmental monitoring records, it was identified that routine weekly surface sampling (RODAC contact plates) and active air sampling for **Cleanroom Suite 114** (Anteroom 114A ISO 8 and Buffer Room 114B ISO 7) were inadvertently omitted by EM analyst SMO for two consecutive weekly monitoring cycles covering **03-Aug-2026 through 14-Aug-2026**, contrary to the requirements of *MICRO-SOP-2 (Environmental Monitoring of the Clean Room Facility)*. The primary root cause was determined to be analyst oversight during high-throughput testing rotations, compounded by the absence of a real-time end-of-week EM schedule reconciliation tracking mechanism.

Because three (3) Celsis sterility testing Out-Of-Specification (OOS) investigations were initiated for sample submissions processed in **Cleanroom 114B** during this exact timeframe, this formal Quality Risk Assessment was conducted to evaluate the state of cleanroom control, analyze batch performance data across 477 concurrent samples, and establish defensible mitigation rationale for each correlated OOS investigation.

### 2. Correlated Sterility Out-Of-Specification (OOS) Investigations

A thorough reconciliation of all testing logs and room assignments in Cleanroom Suite 114 between 03-Aug-2026 and 14-Aug-2026 confirmed three (3) correlated sterility test failure investigations in **Room 114B**:

| Submission ID (ETX) | OOS Number | Testing Room | Client / Sample Description | Test Date | Investigation Status |
|---|---|---|---|---|---|
| **ETX-260729-0660** | **OOS-261824** | **Room 114B** | Boudreaux's New Drug Store (E01607) - Methylcobalamin / Glycine | 03-Aug-2026 | Phase I in progress (Micro ID pending ECD 20-Aug-26) |
| **ETX-260803-0212** | **OOS-261861** | **Room 114B** | Grand Avenue Pharmacy 13409 (E13409) - Triamcinolone Acetonide | 03-Aug-2026 | Phase I in progress (Micro ID ETX-260813-0684 ECD 24-Aug-26) |
| **ETX-260806-0456** | **OOS-261877** | **Room 114B** | Celsis Sterility Batch (Hold as of 17Aug26) | 06-Aug-2026 | Phase I Lab Investigation in progress |

*Note on Excluded Records:* Formal room allocation verification confirmed that **OOS-261878** and **OOS-261932** were processed inside Cleanroom Suite 115 (not Suite 114) and are therefore completely unaffected by this nonconformance and excluded from this evaluation.

### 3. Scientific Risk & Barrier Analysis (State of Environmental Control)

To evaluate whether the missed weekly background EM in Cleanroom Suite 114 contributed to or caused these three (3) sterility test failures, a multi-barrier quality evaluation was conducted:

#### 3.1 Comprehensive Concurrent Batch Performance & Statistical Negative Rate

- **477 Concurrent Samples Audited:** Retrospective audit of all laboratory submissions processed inside Cleanroom Suite 114 across the two-week window (03-Aug-2026 to 14-Aug-2026) identified **477 total samples**.
- **>99.3% Pass Rate / No Widespread Trend:** 474 out of 477 samples successfully passed all sterility acceptance criteria with zero microbial detection. The emergence of only three (3) isolated failures confirms that no widespread, airborne, or systemic bioburden contamination existed in Cleanroom Suite 114 or Room 114B.

#### 3.2 Primary Engineering Control (PEC / ISO 5 BSC) & Session In-Process Monitoring

- **Direct Critical Processing Zone Isolation:** All sample handling, vial manipulations, membrane filtration, and media inoculations were conducted exclusively within certified ISO Class 5 Biological Safety Cabinets (BSCs) located inside Room 114B. The ISO 5 unidirectional vertical laminar airflow provides continuous HEPA barrier protection against background room air.
- **Session-Specific In-Process EM Execution:** Routine session-specific environmental monitoring WAS fully executed for every individual testing process per MICRO-SOP-2 Section 8.1.3:
  - *Passive Air (Settling Plates):* 90 mm TSA plates exposed in the BSC during testing exhibited zero microbial growth (0 CFU/plate, Action Limit ≥1 CFU).
  - *Surface Contact Plates (RODAC):* Work surface contact plates sampled post-testing demonstrated no microbial recovery.
  - *Personnel Monitoring:* Left and right gloved fingertip touch plates (LH/RH) for processing analysts met all alert/action limits.

### 3.3 Continuous Facility Automated Engineering Controls (Lighthouse Monitoring)

- **24/7 Automated Sensor Surveillance:** Physical environmental parameters for Cleanroom Suite 114 (Anteroom 114A and Buffer Room 114B) are continuously monitored and logged electronically by the automated Lighthouse System per SOP 2.700.014.
- **Zero Physical Excursions:** Retrospective audit of Lighthouse continuous data from 03-Aug-2026 through 14-Aug-2026 confirmed continuous positive pressure cascades (ISO 5 BSC -> ISO 7 Buffer Room 114B -> ISO 8 Anteroom 114A -> General Facility) and tight temperature/humidity control (66°F–72°F / 30%–60% RH) with no HVAC shutdowns or pressure reversals.

### 3.4 Cleaning, Disinfection, and Material Transfer Regimen

- Daily and weekly cleaning and sporicidal disinfection protocols (acidified bleach solution and sterile 70% IPA per SOP 2.600.018) were executed and documented according to schedule.
- Staged sanitization (10-minute sporicidal contact time) for all sample bins, consumables, and media bottles prior to cleanroom and BSC introduction remained fully in effect, preventing ingress of external bioburden.

### 3.5 Method Negative Controls & Reagent Verification

- All method negative controls (media blanks, diluent soak blanks, and system rinse controls) processed concurrently with the affected submissions demonstrated valid negative results (baseline RLU on Celsis luminometer), confirming reagent sterility and lack of systemic assay contamination.

### 3.6 Data Gap Mitigation via Microbial Identification & Post-Event Reinstatement

- **Microbial Identification (Micro ID):** For the three OOS cases in Room 114B (OOS-261824, OOS-261861, OOS-261877), recovered microbial isolates are being identified to genus/species to cross-reference against cleanroom baseline flora. Isolates distinct from cleanroom or skin commensals provide definitive evidence of client product container origin.
- **Reinstatement Sampling:** Routine weekly surface and active air EM for Suite 114 was immediately resumed starting 17-Aug-2026, meeting all alert and action limits (Anteroom <50 CFU, Buffer Room <5 CFU), confirming no persistent bioburden accumulation occurred.

## 4. Impact Assessment & Risk Scoring Matrix (Closure by Quality)

| Risk Parameter | Assigned Score | Grading Criteria & Rationale |
|---|---|---|
| **Frequency** | 2 | Between 2 to 10 historical occurrences (isolated supervisory finding across monitoring cycles). |
| **Severity** | 2 | Changes in processes/systems with no retroactive implications; affected batches quarantined under OOS. |
| **Magnitude** | 2 | >2 to <10 Process/Material/Equipment Affected (3 correlated sterility OOS investigations in Room 114B under review). |
| **Overall Grading Score** | **8** | **Score = Frequency (2) × Severity (2) × Magnitude (2) = 8** |
| **Impact Level** | **MAJOR** | **Major Impact Category (Score Range 3 to 17) — appropriately categorized.** |

## 5. Corrective & Preventive Action (CAPA) Summary

1. **Immediate Containment & Reinstatement:** Routine weekly surface and active air EM for Suite 114 was fully reinstated with passing results.
2. **Tracking Sheet Implementation:** Microbiology Supervisor Robin Seymour instituted a mandatory Weekly EM Sampling Tracking Sheet (*EM Monitoring of Cleanrooms Rotation Schedule.xlsx*) requiring weekly reconciliation and supervisory verification.
3. **Personnel Retraining:** Analyst SMO completed documented Read and Understood retraining on MICRO-SOP-2 Section 8.2.
4. **CAR Requirement:** Based on effective root cause remediation and intact environmental barriers, a formal Corrective Action Response (CAR) is **NOT required**.

## 6. Quality Unit Conclusion & Final Disposition

Based on the totality of evidence—including 474 passing concurrent samples (>99.3% pass rate), uninterrupted 24/7 Lighthouse physical parameter stability, satisfactory session-specific ISO 5 BSC settling and contact plate monitoring, acceptable analyst glove touch plates, valid method negative controls, and clean post-reinstatement EM results—the microbiological integrity and physical containment of Cleanroom Suite 114 (114A/114B) remained in a validated state of control during 03-Aug-2026 through 14-Aug-2026. The missed weekly sampling represents an isolated documentation and scheduling oversight without evidence of environmental breakdown. The three (3) correlated sterility OOS investigations in Room 114B remain valid, and each report shall cross-reference NCR# N26436 and incorporate this risk evaluation to support Phase I closure.

| Evaluated / Prepared By: | Reviewed By: | Quality Assurance Approval: |
|---|---|---|
| Qiyue Chen, Ph.D.<br>Project Microbiologist<br>Date: 27-Aug-2026 | Robin Seymour<br>Microbiology Laboratory Supervisor<br>Date: ____________ | Quality Unit Designee<br>Quality Assurance<br>Date: ____________ |